

Homework 1 - CSE 276A - Intro Robotics

Due: Friday, 16 October 2026, before midnight

1. A text file “waypoints.txt” is available on the Canvas website. The structure of the file is individual lines with - x, y, angle. The x, y coordinates are in meters, and the angle is in radians.
2. Design an algorithm to traverse the waypoints. The algorithm should include a kinematic model for the car.
3. Implement the algorithm in python / ROS2 on the Rover/Rubik PI platform.
4. Template code can be found in the repository
https://github.com/AutonomousVehicleLaboratory/rubikpi_ros2/
5. Run the algorithm on the platform with the waypoint file.
6. Provide a report that briefly describes your algorithm, your control algorithm, and your code. Comment on the performance of your algorithm and how you did calibration. Pack report and code into zip file and upload to Canvas.
7. Include a link to an (unlisted) YouTube video with a recording of an example run on your robot in your report. Not the video, but the URL to the video.